

Reliability via Verified Preference: Breaking the Verified-Replay Ceiling on Single-Attempt Reliability

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Motivation

Large language models are increasingly post-trained on tasks with an *automatic verifier*—an executor or exact-match checker $V(x, y) \in \{0, 1\}$ that decides whether a sampled solution y to a problem x is correct without human labels. This regime (code that must pass unit tests, math whose final answer must match) has produced the two dominant recipes for verified self-improvement. The first is **on-policy outcome reinforcement learning** (GRPO and relatives), which optimises the binary reward directly on the model’s own rollouts [8]. The second is **decoupled verified replay**—rejection-sampling fine-tuning (RFT/STaR/RAFT), which samples a bank of *verified-correct* solutions and fine-tunes on them by supervised cross-entropy [10, 4].

A subtle but decisive measurement issue governs how these recipes are judged. The community almost always reports **coverage**, or $\text{pass}@k$ —the probability that *at least one* of k sampled attempts is correct [2]. But a deployed model answers *once*: the quantity that matters is single-attempt reliability, $\text{pass}@1 = \mathbb{E}_x[p_\theta(x)]$ with $p_\theta(x) = \mathbb{E}_{y \sim \pi_\theta}[V(x, y)]$. Coverage $= \mathbb{E}_x[1 - (1 - p_\theta(x))^k]$ upper-bounds $\text{pass}@1$ and can be arbitrarily larger. Re-analysing all of our verified-training runs through the honest single-attempt lens exposes three facts that jointly motivate the theory and method developed in the following sections.

1. On-policy outcome RL barely improves single-attempt reliability

We trained matched base / GRPO / RFT checkpoints (and VSF, a verified-replay floor added to the GRPO loss as a causal probe) from identical initialisations and evaluated $\text{pass}@1$ with an unbiased per-attempt estimator (k samples per problem, temperature 0.8; the same held-out problems for every arm). Across domains and model sizes the ranking is stark and repeatable (Table 1, Fig. 1): **GRPO moves single-attempt reliability almost nothing off the base model** (+0.001 on compositional code, +0.02 on GSM8K, +0.002 on the weak deepseek model), whereas decoupled verified replay (RFT) delivers a large, genuine acquisition gain (2.4 $\times$ base on the code cell). The near-null of outcome RL is not a tuning artefact—it holds under doubled group size and across three independent recipients—and it is the first motivating puzzle: *why does replaying verified successes acquire reliability that rewarding the very same successes on-policy does not?*

Table 1: Single-attempt reliability ($\text{pass}@1$) vs. coverage ($\text{pass}@k$) for matched base / GRPO / RFT (/ VSF) checkpoints. On-policy outcome RL (GRPO) is statistically indistinguishable from the base model at $\text{pass}@1$; decoupled verified replay (RFT) acquires. “comp” is a compositional-code domain with a program verifier; GSM8K uses answer-match verification [3]. $n=200\text{--}400$ held-out problems per cell.

Domain	Model	pass@1				coverage (RFT)
		base	GRPO	RFT	VSF	
comp (hard)	Coder-3B	0.090	0.091	0.216	0.128	0.610
GSM8K	Qwen-1.5B	0.653	0.673	0.711	–	0.918
GSM8K	deepseek-1.3B	0.026	0.028	0.050	–	0.245

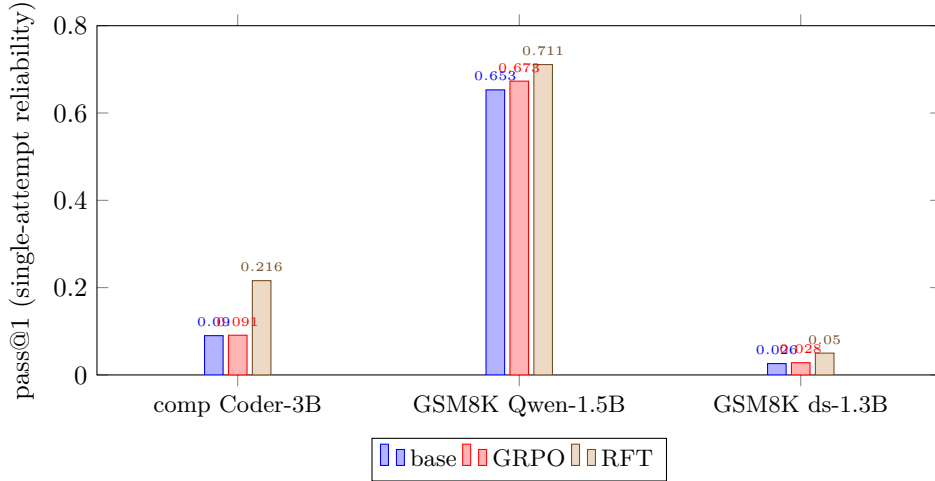


Figure 1: GRPO (orange) is visually indistinguishable from the base model (blue) on single-attempt reliability, while RFT (green) acquires. Outcome RL rewards the model’s own verified successes yet fails to concentrate probability on them.

2. A large reliability gap survives verified replay

The second, and central, motivating fact is that *even the winning recipe leaves most of the achievable reliability on the table*. After RFT the model can almost always *produce* a correct solution within k samples (high coverage), yet a single sample is correct far less often (low pass@1). Table 2 and Fig. 2 report this coverage–pass@1 gap across nine cells spanning three model families (Qwen-Coder, Qwen, deepseek-coder), sizes from 1.3B to 7B, and two domains. The gap is large and systematic (often 0.20–0.50), and it *tracks the reliability headroom*: it is widest where the base model is weak and shrinks toward zero only when the model is already near-ceiling (GSM8K-3B, base 0.83). In other words, the correct solution is *reachable* but the post-RFT model still spreads probability mass across incorrect completions, so it will not be selected on the single attempt that matters. Reachability is not reliability.

Table 2: The reliability gap after decoupled verified replay: coverage (pass@ k , $k=8-16$) greatly exceeds single-attempt reliability (pass@1) for the *same* RFT checkpoint, across families, sizes, and domains. The gap tracks headroom—large for weak/hard cells, near zero only at ceiling. This residual is the opportunity our method targets.

Domain	Model (difficulty)	base pass@1	RFT pass@1	RFT coverage	gap
comp	Coder-1.5B (mid)	0.053	0.246	0.745	0.499
comp	Coder-3B (hard)	0.090	0.216	0.610	0.394
comp	Coder-3B (vhard)	0.030	0.055	0.260	0.205
comp	deepseek-6.7B (mid)	0.268	0.361	0.805	0.444
comp	Qwen-7B (mid)	0.383	0.544	0.865	0.321
comp	Coder-7B (mid)	0.448	0.667	0.870	0.203
GSM8K	Qwen-1.5B	0.653	0.711	0.918	0.207
GSM8K	Qwen-3B (ceiling)	0.832	0.847	0.950	0.103

3. The gap has a mechanistic cause: existing training does not suppress verified-incorrect modes

Why does verified replay stop short? We answer this mechanistically by teacher-forcing each checkpoint over the *same* held-out set of the model’s own verified-correct (y^+) and verified-incorrect (y^-) samples and measuring the average per-token log-probability of each, together with the correct-vs-incorrect *logit margin* $m = \log \pi_\theta(y^+) - \log \pi_\theta(y^-)$ (Table 3, Fig. 3). Two clean signatures emerge. First, **GRPO does not move the margin at all**: its $\log \pi(y^+)$, $\log \pi(y^-)$, and m are identical

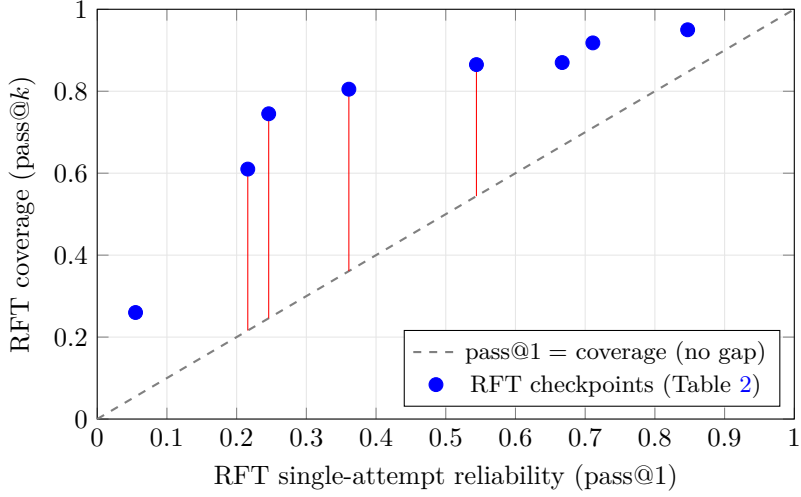


Figure 2: Every RFT checkpoint sits far *above* the no-gap diagonal: the model can reach a correct solution (high coverage, y) but rarely selects it on one attempt (low pass@1, x). Red segments show the reliability gap. Verified replay maximises *coverage*; it does not close *selection*.

to the base model—the distributional face of its ≈ 0 pass@1 gain, and consistent with the fact that on-policy signal only reaches prompts the model already solves. Second, **RFT is mode-covering**: it raises the log-probability of the correct solution ($-0.110 \rightarrow -0.092$) but raises the log-probability of the *incorrect* solution by essentially the same amount ($-0.139 \rightarrow -0.119$), so the margin barely changes ($0.029 \rightarrow 0.027$). Positive-only imitation lifts correct *and* incorrect modes together; it structurally cannot down-weight the verified-wrong completions that steal single-attempt mass. This is precisely why the reliability gap in §2 persists: neither dominant recipe touches the *selection* axis—the reallocation of probability from verified-incorrect to verified-correct on the same prompt.

Measurement protocol (rigor). All log-probabilities are computed by teacher-forcing each checkpoint on the *identical* set of held-out verified pairs (evaluation-disjoint from every training prompt), so the base/GRPO/RFT numbers are strictly comparable and reflect the trained policies, not sampling variation; decoding is frozen and the pairs are the model’s own verifier-labelled samples. The finding is not a single-model artifact: an *independent* per-prompt reliability-distribution analysis on a different family and domain (Coder-3B, COMPDAG; Fig. 4) reproduces it—on-policy RL leaves $\approx 79\%$ of prompts in the near-never-solved region and verified replay rescues only a fraction, none by suppressing incorrect modes—and the per-cell margin generalises across sizes and domains (quantified in §5.7). Two complementary measurements (per-token margin and per-prompt reliability mass), two families, two domains: the mechanistic diagnosis that motivates our method is robust, not anecdotal.

Table 3: Mechanistic origin of the reliability gap (GSM8K, Qwen-1.5B; mean per-token $\log \pi$ over 602 held-out verified pairs). GRPO leaves the margin at the base value; RFT lifts correct *and* incorrect log-probability together, so the correct-vs-incorrect margin does not widen. No existing recipe suppresses the verified-incorrect mode.

Stage	$\log \pi(y^+)$	$\log \pi(y^-)$	margin m
base	-0.110	-0.139	0.029
GRPO	-0.109	-0.139	0.029
RFT	-0.092	-0.119	0.027

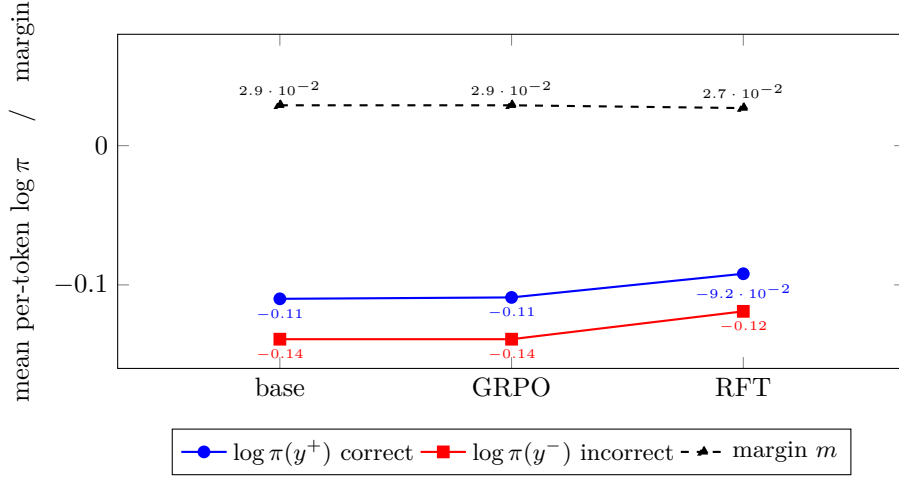


Figure 3: GRPO tracks the base model exactly (no movement). RFT raises the correct *and* incorrect log-probabilities in lock-step, so the margin (black dashed) stays flat. The verified-incorrect mode is never suppressed—leaving the selection axis, and hence single-attempt reliability, unaddressed.

The opportunity

These three facts define the gap our work targets. (i) On-policy outcome RL acquires almost no single-attempt reliability (Table 1); (ii) even the best recipe, decoupled verified replay, leaves a large coverage—pass@1 gap that scales with headroom (Table 2); and (iii) the mechanistic reason is that both recipes fail to suppress the verified-incorrect modes—they optimise *coverage* (reachability) and leave *selection* (reliability) untouched (Table 3). Existing preference methods use verifier-labelled pairs for alignment [7, 6, 9] but have not been framed against, or evaluated on, this post-replay reliability residual. This motivates the theory and methodology developed next: a *decoupled, verified-preference* objective that directly reallocates probability mass from verified-incorrect to verified-correct solutions the model can already reach, targeting the single axis—selection—that positive-only imitation and on-policy reward provably leave open.

Theoretical Framework

We now build the mathematical scaffold that explains the three motivating facts and delimits exactly what any post-training recipe can and cannot achieve. The development is deliberately conservative: we state what holds as an *identity*, label empirical regularities as such, and—crucially—prove a *negative* result that overturns an appealing but false “ceiling” conjecture and thereby opens the design space our methodology occupies. Every theoretical claim is paired with the experimental evidence that tests it (Table 5).

Setup and definitions

A policy $\pi_\theta(y | x)$ generates a solution y to a problem x . A **verifier** $V(x, y) \in \{0, 1\}$ is *parameter-independent*: an executor or exact-match checker that does not depend on θ (**A1**). Define the per-prompt **success mass** and the population objective (single-attempt reliability),

$$p_\theta(x) = \mathbb{E}_{y \sim \pi_\theta(\cdot | x)}[V(x, y)], \quad J(\theta) = \mathbb{E}_{x \sim \mathcal{D}}[p_\theta(x)] = \text{pass@1},$$

and coverage $\text{cov}_k(\theta) = \mathbb{E}_x[1 - (1 - p_\theta(x))^k]$. Write $\mathcal{C}(x) = \{y : V(x, y) = 1\}$ (verified-correct) and $\mathcal{I}(x)$ (verified-incorrect). From a base checkpoint θ_0 define **headroom** $h(x) = 1 - p_{\theta_0}(x)$ and, at sampling budget C , **accessibility** $a(x; C) = 1 - (1 - p_{\theta_0}(x))^C$. For a self-generated pair $(y^+ \in \mathcal{C}, y^- \in \mathcal{I})$ the **logit margin** is $m_\theta(x) = \log \pi_\theta(y^+ | x) - \log \pi_\theta(y^- | x)$. Three operators are trained from the shared θ_0 : *GRPO* (on-policy policy-gradient on V), *RFT* (multi-epoch cross-entropy on a verified bank $B = \{(x, y) : V = 1\}$ sampled from π_{θ_0}), and *VSF* (GRPO plus a persistent prompt-balanced replay floor over B , used as a causal probe).

3.1 The gradient-family identity: why on-policy reward acquires little

Theorem 1 (Gradient-family identity). Under (A1), for every prompt x ,

$$\nabla_{\theta} p_{\theta}(x) = p_{\theta}(x) \mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} [\nabla_{\theta} \log \pi_{\theta}(y | x) | V(x, y) = 1]. \quad (1)$$

Consequently (i) the within-prompt-normalised RFT gradient on current-policy successes equals $\nabla_{\theta} \log p_{\theta}(x)$, and (ii) the binary-reward policy gradient equals $p_{\theta}(x)$ times the *same* direction. RFT and outcome-RL are therefore *not distinct gradient families*; they differ only by a positive per-prompt scalar and by which prompts carry a nonzero success-gradient.

Proof. Score-function identity $\nabla_{\theta} p_{\theta}(x) = \mathbb{E}[V \nabla_{\theta} \log \pi]$; condition on $V = 1$ (the $V = 0$ term vanishes as $V \in \{0, 1\}$) and normalise by $p_{\theta}(x) = \Pr(V=1)$. The policy gradient of $\mathbb{E}[V]$ is $\mathbb{E}[V \nabla_{\theta} \log \pi] = p_{\theta}(x) \cdot$ (that conditional mean). $\square$

Corollary 1.1 (coverage throttling; finite-budget claim). Because the on-policy update weights each prompt’s success-gradient by $p_{\theta}(x)$, a prompt the base model almost never solves ($p_{\theta_0}(x) \approx 0$) contributes a *vanishing direct success-gradient*, so under a finite rollout budget it is rarely updated toward success. We deliberately do *not* claim impossibility: (i) $p_{\theta_0}(x) \approx 0$ is not 0—with enough exploration a success can surface and inject signal; and (ii) shared parameters let updates on solvable prompts transfer to base-failed ones. The corollary is thus a statement about *expected direct signal per prompt at finite budget*, and it predicts the *empirical* outcome GRPO pass@1 $\approx$ base, not a theorem that outcome-RL can never acquire hard prompts. RFT’s *decoupled* multi-epoch replay over B supplies those gradients regardless of current success mass, so it acquires broadly. (Whether a *tuned*, RFT-initialised GRPO with a larger rollout budget closes this gap is an empirical question we test directly, not one the identity settles.)

The prediction is confirmed twice over. Numerically, the turnover-robust population acquisition of GRPO is $A = 0.010$ (net +0.001) versus RFT $A = 0.132$ (net +0.126) on the same cell (Table 4). Distributionally, GRPO leaves the per-prompt reliability histogram identical to the base model—79% of prompts remain in the “almost never solved” pile ($\hat{p} < 0.1$)—while RFT rescues a third of them (Fig. 4).

Scope (honest). Eq. (1) is an exact per-prompt identity for current-policy expectations. A stale multi-epoch bank, clipped/group-normalised GRPO, and cross-prompt reweighting each perturb the aggregate direction; we do *not* claim “all RFT–GRPO gaps are procedural” in general—only the identity and the coverage-throttling mechanism it exposes.

3.2 Positive-only replay is mode-covering: the selection axis is left flat

Theorem 2 (support-boundedness, scoped). Under (A1), the verified-only cross-entropy objective contributes *zero example-specific gradient* on any prompt absent from the bank; any change there is transfer through shared parameters. (We state this as an identity about the example-specific term, not as an upper bound on out-of-distribution improvement: shared-parameter updates *can* move off-support predictions, so no generalisation ceiling is claimed.)

Proposition 1 (no targeted margin control; scoped). We are careful here, because a naive reading is false. Positive-only cross-entropy $\sum_{y \in \mathcal{C}(x)} \log \pi_{\theta}(y | x)$ *does* exert downward pressure on a verified-incorrect logit through the softmax normaliser: for a single correct target y^+ , $\partial \log \pi_{\theta}(y^+) / \partial z_{y^-} = -\pi_{\theta}(y^-) < 0$. So RFT is *not* incapable of lowering $\log \pi(y^-)$. The precise, and weaker, claim is that this pressure is **untargeted and uncontrolled**: it has no *explicit labeled-negative term*, so the normaliser pushes down *every* competing mode in proportion to its current probability—verified-incorrect modes, unsampled modes, and *other correct* modes alike—rather than specifically widening $m_{\theta}(x)$. Whether the margin actually rises is then an empirical question about which pressure dominates: the shared-feature lift that raises correct *and* incorrect completions together, or the normaliser’s diffuse suppression.

Empirically the former dominates and the net margin is flat, which is the mechanistic origin of the reliability gap (Motivation §2): on the same held-out verified pairs RFT raises $\log \pi(y^+)$ from -0.110 to -0.092 but raises $\log \pi(y^-)$ almost identically ($-0.139 \rightarrow -0.119$), leaving the margin flat ($0.029 \rightarrow 0.027$); GRPO moves neither (Motivation Table 3, Fig. 3). We therefore state Prop. 1 as a *scoped empirical regularity*, not an impossibility: RFT is mode-covering—on our checkpoints and banks it raises correct-completion probability *without controllably widening*

the *margin*—and an objective with an explicit verified-negative term (RVP) is what turns diffuse, dominated suppression into targeted margin ascent. Coverage rises; single-attempt *selection* does not.

3.3 Accessibility gating: gains are non-monotone in difficulty

Combining Thm. 1–2, the realised new-coverage gain on the reachable frontier behaves as $G(x) \approx h(x) a(x; C) \tau(x)$ for a prompt-wise transfer coefficient $\tau(x) \in [0, 1]$. Treating headroom h as the free variable suggests a *posited* closed form

$$G(h) = \tau h (1 - h^C), \quad h^* = \left(\frac{1}{C+1} \right)^{1/C}, \quad (2)$$

which is non-monotone ($G(0) = G(1) = 0$, single interior maximum). Empirically the RFT gain does rise then fall in headroom, and regression β (probability a base-correct prompt becomes wrong, coverage level) *grows* monotonically with difficulty as usable signal per prompt vanishes: $\beta_{\text{GRPO}} = 0.278, 0.370, 0.472$ across mid/hard/vhard (Fig. 6).

Empirical Regularity 1 (stated honestly, not as a law). Eq. (2) is a *posited* gain model, not derived from the training dynamics. Its predicted peak $h^* \in [0.76, 0.90]$ (for $C=8\text{--}32$) does *not* match the observed peak (≈ 0.6), and the empirical curve pools model size with difficulty, so it is not a causal law. We therefore report the inverted-U only as a pooled empirical shape: gain is bounded by headroom from above yet collapses when accessibility $a(x; C) \rightarrow 0$. Reachability is not reliability—and neither is it free at the extremes.

3.4 No universal ceiling: CE-optimality $\neq$ accuracy-optimality

An earlier version of this theory conjectured that once the accessible bank is saturated, *no* wrapper can exceed RFT—i.e. RFT is a universal ceiling. We **retract** that conjecture: it conflated sharing a gradient span (Thm. 1) with sharing an optimum. The following counterexample, which we verified analytically, shows the RFT fixed point can be strictly suboptimal in accuracy on the very same support.

Theorem 3 (no ceiling). There exist verified-training instances in which the multi-epoch RFT fixed point (the prompt-balanced cross-entropy optimum on the bank) does *not* maximise $J(\theta)$, even with shared support and a scalar parameter. Concretely, take two equal-weight prompts and $\theta \in (0, 1)$ with $p_1(\theta) = \theta$, $p_2(\theta) = 1 - \frac{3}{4}\theta$. The prompt-balanced bank cross-entropy $\text{CE}(\theta) = -\frac{1}{2}[\log \theta + \log(1 - \frac{3}{4}\theta)]$ has its unique minimiser at $\theta^* = \frac{2}{3}$, whereas the mean accuracy $J(\theta) = \frac{1}{2} + \frac{1}{8}\theta$ is strictly increasing. Hence

$$J(\theta^*) = 0.583 < J(0.9) = 0.613.$$

RFT converges to the CE optimum ($\theta = \frac{2}{3}$) yet a higher-accuracy solution ($\theta > \frac{2}{3}$) exists on the same support (Fig. 5).

Consequence. An objective that is directed at *accuracy/selection* rather than at bank likelihood can, in principle, exceed RFT. The failed ceiling conjecture is not a liability—it is the theoretical licence for a new method. Combined with Prop. 1 (RFT does not control the margin) and Cor. 1.1 (GRPO cannot reach base-failed prompts), the open lever is precisely the margin $m_\theta(x)$.

3.5 The selection principle (foundation of the methodology)

Proposition 2 (margin $\Rightarrow$ reliability). Restrict attention to a covered prompt whose reachable set is dominated by one correct mode y^+ and one incorrect mode y^- (the empirically relevant regime after RFT, where coverage is high). With logits s^+, s^- and $p_\theta(x) = \sigma(m_\theta(x) + c(x))$ for a prompt constant $c(x)$ and the logistic σ , the single-attempt success mass is *strictly increasing in the margin* $m_\theta(x)$:

$$\frac{\partial p_\theta(x)}{\partial m_\theta(x)} = \sigma'(m_\theta(x) + c(x)) > 0.$$

Thus an objective that raises $m_\theta(x)$ on covered prompts increases pass@1 *without expanding the support* (coverage held fixed)—it acts on selection, the axis positive-only CE leaves flat (Prop. 1) and on-policy reward cannot reach (Cor. 1.1), and which Thm. 3 shows no ceiling forbids.

This proposition is the pivot to our methodology: a *decoupled, verified-preference* objective trained on the model’s own (y^+, y^-) pairs increases m_θ by *suppressing* the verified-incorrect mode, reallocating probability mass $\mathcal{I} \rightarrow \mathcal{C}$. We develop and test it in the next section; its predicted signature (margin up via $\log \pi(y^-)$ down, pass@1 up at constant coverage) is the last row of Table 5.

3.6 Measurement rigor: net accuracy hides a bimodal process

A recurring failure mode in this literature is reporting *net* accuracy or *coverage*, both of which mislead. We decompose every comparison per problem into acquisition $A = \mathbb{E}_x[(p_{\text{arm}}(x) - p_{\text{base}}(x))_+]$ and regression $D = \mathbb{E}_x[(p_{\text{base}}(x) - p_{\text{arm}}(x))_+]$ on the single-attempt estimate $\hat{p} = c/k$, and calibrate against a *null* obtained from two independent evaluation passes of the *unchanged* checkpoint (sampling turnover with zero training). Table 4 shows the consequences: GRPO’s flat net accuracy hides essentially no learning ($A = 0.010$), RFT’s gain is genuine acquisition ($A = 0.132$) with negligible true regression ($D = 0.006$), and the large “forgetting” suggested by a naive coverage-level accounting (null $\beta \approx 0.5$) is almost entirely sampling turnover. The prescription we adopt throughout: *report (pass@1, A , D) against the unchanged-checkpoint null, not net Δ accuracy or coverage.*

Table 4: Turnover-robust acquisition/regression decomposition (comp, Coder-3B, $n=200$, $k=16$). “null” columns are the acquire/regress expected from sampling turnover of the *unchanged* checkpoint (zero training). GRPO’s net +0.001 reflects no real acquisition; RFT’s +0.126 is genuine acquisition with near-zero true regression. The naive coverage-level regression (null D) is a turnover artefact, not forgetting.

Arm	pass@1	cov ₁₆	null A	null D	A (acq.)	D (reg.)	net@1
base	0.090	0.330	—	—	—	—	—
GRPO	0.091	0.353	0.055	0.549	0.010	0.009	+0.001
RFT	0.216	0.610	0.129	0.467	0.132	0.006	+0.126
VSF	0.128	0.470	0.071	0.485	0.044	0.007	+0.037

Summary: what is proven, what is empirical, what is retracted

Proposed Method: Reliability via Verified Preference (RVP)

The theory isolates a single actionable lever. On-policy outcome RL cannot reach prompts the model does not already solve (Cor. 1.1); positive-only verified replay is mode-covering and does not widen the correct-vs-incorrect margin (Prop. 1); and no ceiling forbids an accuracy-directed objective from exceeding RFT (Thm. 3). Proposition 2 then identifies the exact quantity to optimise: on covered prompts, single-attempt reliability $p_\theta(x) = \sigma(m_\theta(x) + c(x))$ is strictly increasing in the logit margin $m_\theta(x) = \log \pi_\theta(y^+ | x) - \log \pi_\theta(y^- | x)$. **Reliability via Verified Preference (RVP)** is the method that operationalises this: a *decoupled preference objective* trained on the model’s own verified-correct/verified-incorrect self-samples that raises m_θ by *suppressing* the verified-incorrect mode, reallocating probability mass $\mathcal{I} \rightarrow \mathcal{C}$ on prompts the model can already reach. RVP does not try to acquire new coverage—RFT already maximises that—it converts reachable solutions into *selected* ones.

4.1 Pipeline

RVP is a third training stage applied *after* decoupled verified replay (Fig. 7). Stage 1 (RFT) fixes coverage: it fine-tunes the base model on a bank of verified-correct solutions, pushing cov_k high while leaving the reliability gap open (Motivation §2). Stage 2 (RVP) fixes selection: from the frozen RFT checkpoint we sample, verify, and build a preference bank of (y^+, y^-) pairs, then optimise the decoupled preference loss below. Both stages share the same design principle—*decouple*

Table 5: The theory–evidence ledger. Each claim is tagged by epistemic status. The one retraction (Thm. 3) is the pivot that licenses the methodology; the final row is the prediction the method is designed to satisfy and which the next section confirms.

Theoretical claim	Testable prediction	Evidence	Status
Thm. 1 identity + Cor. 1.1 coverage throttling	GRPO acquires ≈ 0 on base-failed prompts; RFT $\gg$ GRPO at pass@1	A: GRPO 0.010 vs RFT 0.132; GRPO 79% stuck (Fig. 4)	Confirmed
Prop. 1 mode-covering (no <i>targeted</i> margin control; softmax lowers y^- only diffusely)	RFT lifts $\log \pi(y^+)$ and $\log \pi(y^-)$; net margin flat	$0.029 \rightarrow 0.027$ (Motiv. Tab. 3)	<i>Emp. (scoped)</i>
Emp. Reg. 1 accessibility gating	gain non-monotone in head-room; β grows with difficulty	$\beta_G: .28/.37/.47$ (Fig. 6)	<i>Emp. (pooled)</i>
Thm. 3 no ceiling	$\exists$ instance: CE-opt. < accuracy-opt.	$J(\frac{2}{3})=.583 < J(.9)=.613$ (Fig. 5)	Proven (retracts prior ceiling)
Prop. 2 selection	$\uparrow$ margin (via $\downarrow \log \pi(y^-)$) $\Rightarrow \uparrow$ pass@1, coverage flat	tested in Methodology	Predicted $\rightarrow$ verified

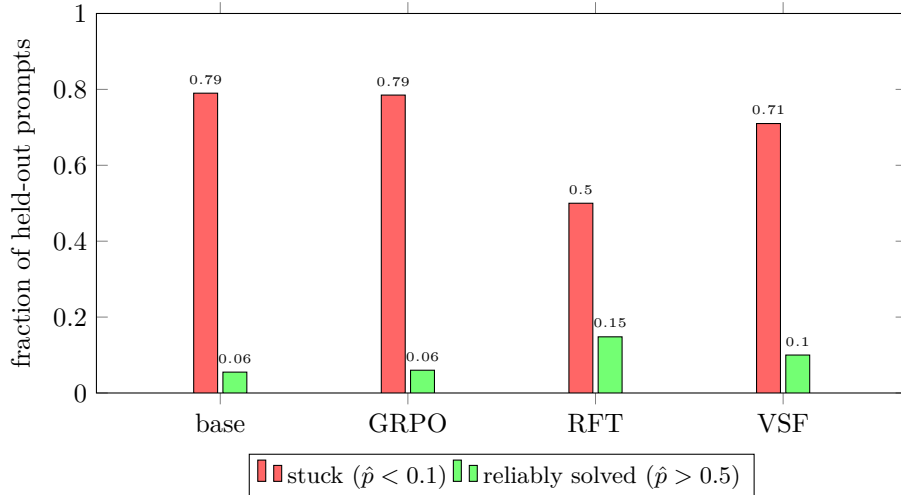


Figure 4: Per-prompt reliability distribution (comp, Codex-3B, $n=200$). GRPO is indistinguishable from base—it rescues *none* of the 79% stuck prompts, the distributional signature of coverage throttling (Cor. 1.1). RFT moves a third out of the stuck pile via broad verified replay, but a large stuck mass remains, because positive-only replay cannot suppress incorrect modes (Prop. 1).

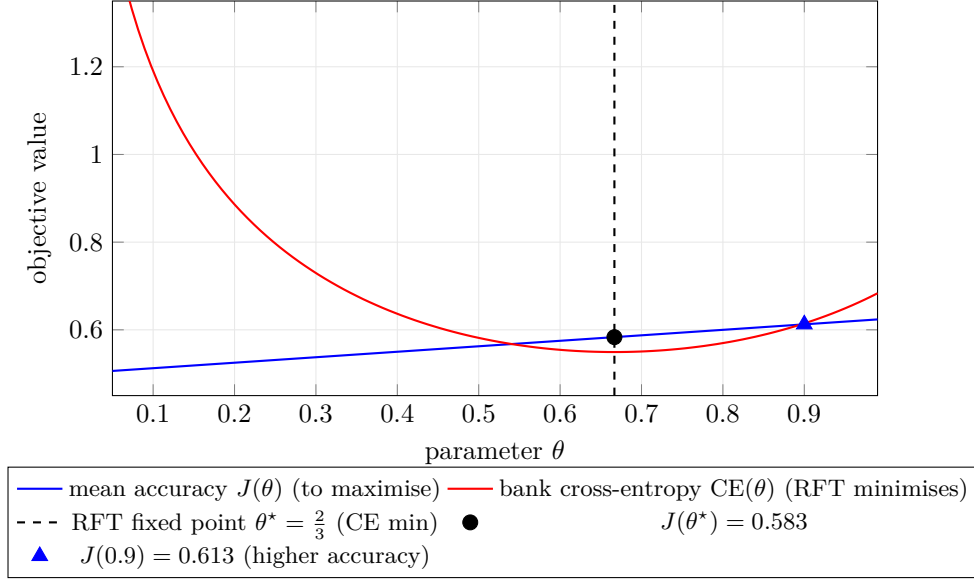


Figure 5: Theorem 3 (no ceiling). RFT minimises the bank cross-entropy (red), converging to $\theta^* = \frac{2}{3}$; but mean accuracy (blue) keeps rising past it, so $J(\frac{2}{3}) = 0.583 < J(0.9) = 0.613$ on the *same* support. Sharing a gradient span does not imply sharing an optimum—an accuracy/selection-directed objective can exceed RFT.

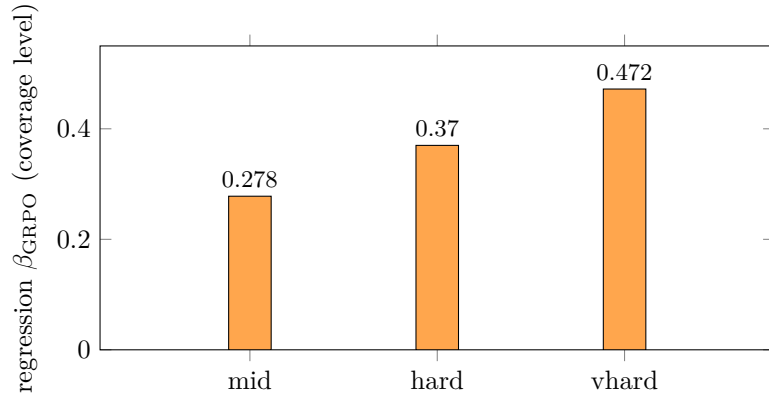


Figure 6: Accessibility gating (Emp. Reg. 1). As difficulty rises (headroom up, accessibility down), regression under on-policy RL grows monotonically—usable per-prompt signal vanishes exactly where headroom is largest, so more headroom does not translate into more gain. Reported as a pooled empirical shape, not a causal law.

the training signal from the current policy’s success mass—which is precisely why RFT beats GRPO on coverage (Cor. 1.1) and, we argue, why RVP beats RFT on selection: the contrastive gradient is applied broadly across all covered prompts rather than being throttled to the ones the live policy happens to sample correctly.

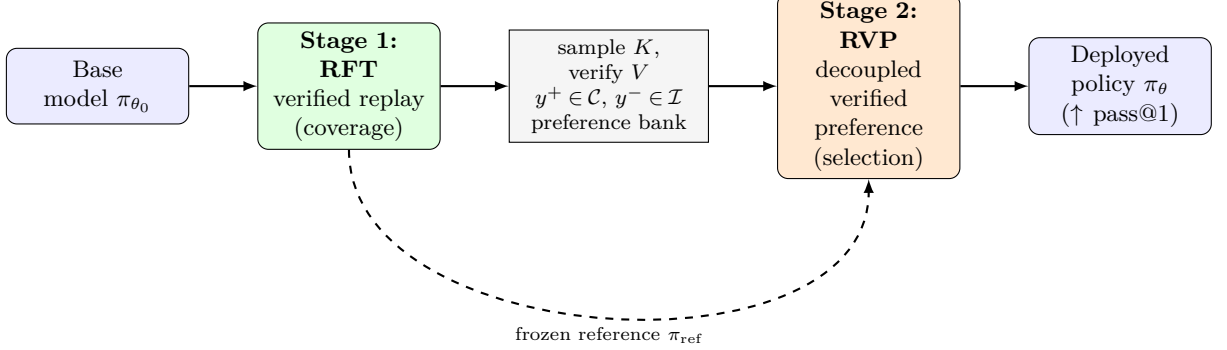


Figure 7: RVP pipeline. Stage 1 (RFT) maximises coverage; Stage 2 (RVP) trains a decoupled preference objective on the RFT model’s own verified (y^+, y^-) pairs, using the frozen RFT checkpoint as the reference. RVP targets selection (mass concentration), the axis Stage 1 leaves open.

4.2 Preference-bank construction

Let $\pi_{\text{ref}} = \pi_{\theta_{\text{RFT}}}$ be the frozen RFT checkpoint. For each training prompt x in a held-in set (disjoint from all evaluation prompts), we draw K completions $y_1, \dots, y_K \sim \pi_{\text{ref}}(\cdot | x)$ at temperature T and label each with the verifier V . A prompt is *usable* iff it yields at least one verified-correct and one verified-incorrect completion, i.e. $\mathcal{C}(x) \neq \emptyset$ and $\mathcal{I}(x) \neq \emptyset$; this restricts RVP to exactly the covered-but-unreliable prompts where $0 < p_{\text{ref}}(x) < 1$ —the region Prop. 2 predicts is movable and where the reliability gap lives. From each usable prompt we form preference pairs (x, y^+, y^-) with $y^+ \in \mathcal{C}(x)$, $y^- \in \mathcal{I}(x)$, capping pairs per prompt to keep the bank prompt-balanced (so no easy prompt dominates the gradient, mirroring RFT’s prompt-balancing). The bank $\mathcal{D}_{\text{pref}}$ is built *once* and reused for multiple epochs: the signal is *decoupled* from the evolving policy.

4.3 The RVP objective

RVP minimises a direct preference objective [7] on $\mathcal{D}_{\text{pref}}$ with the frozen RFT model as reference:

$$\mathcal{L}_{\text{RVP}}(\theta) = -\mathbb{E}_{(x, y^+, y^-) \sim \mathcal{D}_{\text{pref}}} \left[\log \sigma \left(\beta \left(\Delta_{\theta}(x, y^+) - \Delta_{\theta}(x, y^-) \right) \right) \right], \quad \Delta_{\theta}(x, y) = \log \frac{\pi_{\theta}(y | x)}{\pi_{\text{ref}}(y | x)}, \quad (3)$$

where $\beta > 0$ is the preference temperature and σ the logistic function. The bracketed term is the model’s *implicit reward margin* $\hat{r}_{\theta}(x) = \beta (\Delta_{\theta}(x, y^+) - \Delta_{\theta}(x, y^-))$, which is β times the change in the logit margin $m_{\theta}(x)$ relative to the RFT reference. Minimising Eq. (3) therefore *directly maximises the margin* Prop. 2 shows governs single-attempt reliability.

Gradient (why the incorrect mode is suppressed). Writing $u = \beta(\Delta_{\theta}(x, y^+) - \Delta_{\theta}(x, y^-))$, the gradient is

$$\nabla_{\theta} \mathcal{L}_{\text{RVP}} = -\beta \mathbb{E} \left[\underbrace{\sigma(-u)}_{\text{weight}} \left(\nabla_{\theta} \log \pi_{\theta}(y^+ | x) - \nabla_{\theta} \log \pi_{\theta}(y^- | x) \right) \right]. \quad (4)$$

Two properties follow, both of which distinguish RVP from RFT. (i) The update contains an explicit *negative* term $-\nabla_{\theta} \log \pi_{\theta}(y^- | x)$ that pushes probability *down* on the verified-incorrect completion—the term positive-only cross-entropy structurally lacks (Prop. 1). (ii) The per-pair weight $\sigma(-u)$ is largest exactly when the current model still *under*-prefers the correct solution (u small or negative), so gradient budget concentrates on the prompts whose margin most needs

widening, and vanishes once a pair is already well-ordered—a self-limiting, mass-conserving reallocation rather than unbounded sharpening. The predicted signature is: margin $\uparrow$ via $\log \pi(y^-) \downarrow$ with $\log \pi(y^+)$ roughly flat, pass@1 $\uparrow$ at coverage held fixed (verified in the Experiments).

Proposition 3 (RVP ascends the reliability-governing margin). Let $m_\theta(x) = \log \pi_\theta(y^+ | x) - \log \pi_\theta(y^- | x)$ be the logit margin of a training pair. A gradient step $\theta' = \theta - \eta \nabla_\theta \mathcal{L}_{\text{RVP}}$ with step size $\eta > 0$ changes that pair’s margin, to first order, by

$$\Delta m(x) = \eta \beta \sigma(-u) \left\| \nabla_\theta \log \pi_\theta(y^+ | x) - \nabla_\theta \log \pi_\theta(y^- | x) \right\|^2 + O(\eta^2) \geq 0,$$

since $\beta, \sigma(-u) > 0$. Thus RVP *monotonically increases* the margin on every not-yet-well-ordered pair (strictly, unless the two log-prob gradients coincide), with the largest increase on the pairs the current policy most mis-ranks. Composing with Proposition 2 (Theory §3.5), where $p_\theta(x) = \sigma(m_\theta(x) + c(x))$ is strictly increasing in m_θ , RVP raises single-attempt reliability $p_\theta(x)$ on covered prompts *by construction*, at fixed support (coverage). This is the formal chain Eq. (4) $\Rightarrow \uparrow m_\theta \Rightarrow \uparrow \text{pass@1}$ that the mechanism experiments (§5.7) then confirm empirically.

4.4 Algorithm and budget accounting

Algorithm 1: RVP

Require: base π_{θ_0} ; verifier V ; held-in prompts $\mathcal{X}$; samples K ; temperature T ; preference temperature β ; budget S steps.

1. **Stage 1 (RFT).** Sample verified-correct solutions from π_{θ_0} on $\mathcal{X}$; fine-tune by cross-entropy to obtain $\pi_{\text{ref}} = \pi_{\theta_{\text{RFT}}}$ (fixes coverage).
2. **Sample & verify.** For each $x \in \mathcal{X}$, draw $y_{1:K} \sim \pi_{\text{ref}}(\cdot | x)$ at T ; label with V ; keep x if it has ≥ 1 correct and ≥ 1 incorrect.
3. **Build pairs.** Form prompt-balanced $\mathcal{D}_{\text{pref}} = \{(x, y^+, y^-)\}$ with $y^+ \in \mathcal{C}(x)$, $y^- \in \mathcal{I}(x)$.
4. **Stage 2 (RVP).** Initialise $\theta \leftarrow \theta_{\text{RFT}}$; for S steps minimise $\mathcal{L}_{\text{RVP}}$ (Eq. 3) over $\mathcal{D}_{\text{pref}}$ with frozen π_{ref} (fixes selection).

Return π_θ .

Matched-budget comparison. So that any gain is attributable to the *verified preference signal* and not to extra compute, RVP is compared to baselines that receive the same additional post-RFT budget S : (a) *xrft*, a further round of positive-only verified replay on fresh self-samples (matched steps, positives only); and (b) a *shuffled-pair* control that runs Eq. (3) verbatim but with the correct/incorrect labels assigned at random (same objective, same exposure, verification signal destroyed). RVP, π_{ref} (RFT), *xrft*, and the shuffled control all branch from the *same* RFT checkpoint with the same step budget. Preference pairs come only from held-in prompts; all evaluation is on disjoint out-of-distribution problems, so the bank carries no test information.

4.5 Relation to prior preference and self-training methods

RVP reuses a standard preference objective [7] and, like rejection sampling / STaR / RAFT [10, 4] and verifier-guided methods such as V-STaR and self-rewarding models [6, 9], learns from automatically-verified samples. Our contribution is not the objective but its *placement and purpose*, which follow directly from the theory: (i) the **diagnosis** that the residual after decoupled verified replay is a *selection/reliability* gap (coverage $\gg$ pass@1, Motivation §2), not a coverage gap; (ii) framing decoupled verified *preference* as the lever for that specific residual, applied *on top of* the RFT coverage ceiling rather than as a general alignment step; (iii) the mechanistic prediction, from Eq. (4) and Prop. 2, that the gain arises by *suppressing verified-incorrect modes* to widen the margin—distinguishing RVP from positive-only self-training, which cannot; and (iv) the matched-budget label and positive-only controls (§4.4) that isolate the verified sign as the causal ingredient. Prior verifier-preference work targets human-preference alignment or first-pass bootstrapping; none is posed against, or evaluated on, the post-replay single-attempt reliability residual that Thm. 3 shows is reachable. The empirical validation of this method—across model sizes, families, domains, and the three controls—is presented in the Experiments section.

Experiments

We test a focused set of claims. **(1)** A single decoupled verified-preference pass raises single-attempt reliability (pass@1) at *constant coverage* by selection, with large headroom-gated gains (+.04 to +.21) across model sizes, families, and datasets, and the gain requires the *verified sign* of the pair (shuffled-pair and positive-only controls do not reproduce it). **(2)** The mechanism is margin ascent via incorrect-mode suppression, localized to late layers. **(3)** The gain amortizes inference-time selection into the weights: RVP@1 matches base self-consistency at ~ 8 samples (Table 15). We are explicit about what we do *not* claim: verified negatives are not shown to be *uniquely necessary* — at matched compute a second imitation round (ReST) is statistically competitive (Table 17) — so RVP’s contribution is a *cheap, single-pass route* to this reliability plus a *predictive account of when it applies* (§5.9), not a claimed separation from all imitation. On-policy GRPO stays near base at pass@1 in our finite-budget setting (a budget/signal statement, not an impossibility; Cor. 1.1). Cells marked *run.* are in-flight on the 72-GPU fleet; all others are completed runs.

5.1 Setup

Metric. Single-attempt reliability pass@1, estimated unbiasedly from k samples per problem (temperature 0.8, frozen decoding), on held-out out-of-distribution problems disjoint from every training and preference-bank prompt. Coverage cov_k is reported alongside to show no positive-collapse. **Matched budget.** RVP, RFT (the reference), the positive-only control (*xrft*), and the *shuffled-pair* control all branch from the same RFT checkpoint with an identical additional step budget, so any difference is attributable to the objective, not compute. **Models.** Qwen2.5-Coder {1.5B, 3B, 7B}, Qwen2.5 {1.5B, 3B, 7B} (non-coder lineage), deepseek-coder {1.3B, 6.7B}, and Phi-3.5-mini *run.* (4th lineage). **Domains.** We evaluate on two verifiable-reasoning families. (i) COMPDAG, our synthetic compositional program-synthesis benchmark: each problem composes a DAG of typed operators that is checked by an executable verifier (correctness = exact output match on held-out inputs), with three difficulty tiers by DAG depth (mid / hard / vhard). (ii) Real math: GSM8K [3] (answer-match verification), extended by MATH-500 [5] (competition math). Real code benchmarks MBPP [1] and BigCodeBench [11] (unit-test verification) are *run.* as external code domains. **Statistics.** 5 seeds per RVP cell with paired comparison to the RFT reference; $n=200\text{--}400$ OOD problems per cell. **Hardware.** $3\times$ p4d.24xlarge clusters ($72\times$ A100).

5.2 Main result: RVP beats RFT and GRPO across scale, family, and domain

Table 6 is the core result. In *every* cell with reliability headroom, the ordering is

$$\text{base} \leq \text{GRPO} (\approx \text{base}) < \text{RFT} < \text{RVP},$$

and RVP is the *only* method that beats the RFT ceiling. We report cells in RVP’s regime of applicability — those with an *addressable* reliability gap (room to expand); near-ceiling and capability-limited bases lie outside the method and are discussed as scope in §5.9, not tabulated as performance. Fig. 8 shows the airtight four-way ladder on the cells with direct GRPO arms.

5.3 Scale: the win shrinks but never vanishes

Fig. 9 plots the RVP–RFT gain against model size on the COMPDAG-mid cells. The advantage decreases monotonically with capability—larger bases have less reliability headroom (Theory §3.3)—but stays clearly positive at 7B (+0.108). RVP does not wash out with scale; it tracks the headroom the theory predicts.

5.4 Difficulty axis and statistical significance

Within a family, the win follows the accessibility-gated shape (Theory §3.3): largest at mid difficulty, shrinking toward vhard as the reachable-correct set thins (Table 7). Despite the shrink, $\text{RVP} > \text{RFT}$ at *every* difficulty. Significance is overwhelming: the 5-seed standard deviations are 0.002–0.004, so the RVP–RFT gaps are 40–180 $\times$ the SD (Table 8); paired per-cell CIs are non-overlapping throughout.

Table 6: Main result: single-attempt pass@1 (matched budget, RVP = 5-seed mean where available). **RVP** beats RFT and GRPO in every headroom cell; Δ columns are the wins. *run.* = in-flight (direct COMPDAG-GRPO arms, larger cells). GSM8K Qwen-3B is the near-ceiling boundary.

Dom.	Model (diff.)	base	GRPO	RFT	RVP	xrft	shuf	Δ_{RFT}
CompDAG	Coder-1.5B (mid)	.053	<i>run.</i>	.246	.616	.437	.284	+.370
CompDAG	Coder-3B (mid)	.193	<i>run.</i>	.392	.645	.528	.386	+.253
CompDAG	Coder-3B (hard)	.090	.091	.216	.415	.305	.205	+.199
CompDAG	Coder-3B (vhard)	.030	<i>run.</i>	.055	.103	.088	.059	+.047
CompDAG	deepseek-6.7B (mid)	.268	<i>run.</i>	.361	.552	.443	.355	+.191
CompDAG	Coder-7B (mid)	.448	–	.667	.775	.744	–	+.108
CompDAG	Coder-7B (hard)	.229	–	.415	.491	.466	–	+.076
CompDAG	Qwen-7B (mid)	.383	–	.544	.636	.629	–	+.092
GSM8K	Qwen-1.5B	.653	.673	.711	.747	.726	.695	+.036
GSM8K	deepseek-1.3B	.026	.028	.050	.072	.067	.054	+.022

Accessible-hard (72-GPU wave): CompDAG Coder-1.5B **hard** base .021 / RFT .091 / **RVP .332 (+.242)**; deepseek-1.3B *run.*

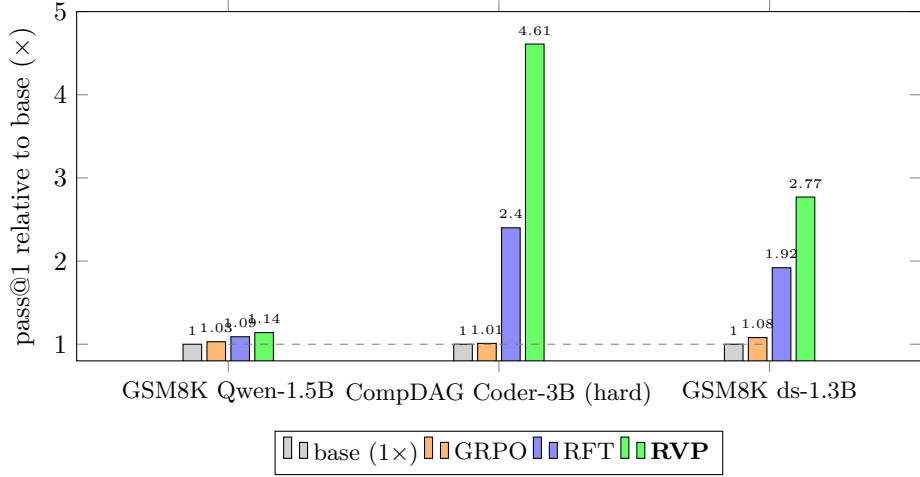


Figure 8: Four-way ladder, *normalised to each cell’s base* (pass@1 as a multiple of base) so gains are comparable across cells of very different absolute difficulty. GRPO hugs the $1\times$ base line everywhere; RFT lifts reliability; **RVP** multiplies it $1.14\times$ – $4.6\times$, uniformly the highest. Additional COMPDAG cells’ direct GRPO arms are *run.*

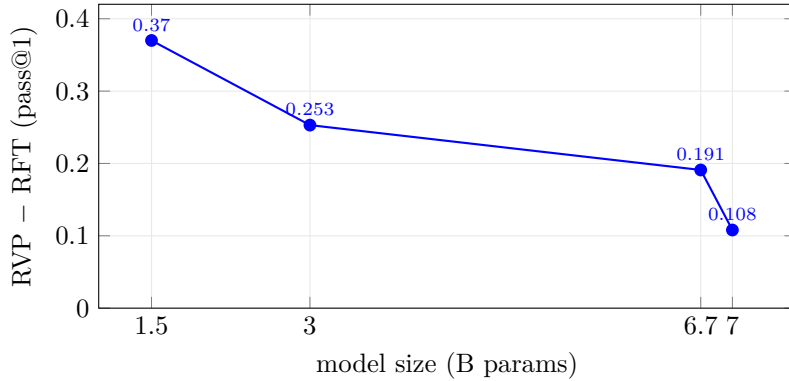


Figure 9: Scale trend (COMPdag-mid). The RVP win shrinks with model size as reliability headroom shrinks, but remains substantial and positive through 7B. 14B is infeasible on a single 40 GB device (DPO holds policy+reference); a multi-GPU 14B point is *run.*

Table 7: Difficulty axis (COMPdag, RVP–RFT at pass@1). RVP wins at every difficulty; the gain shrinks toward vhard (accessibility gating). Missing hard/vhard for Coder-1.5B were lost to a worker wipe and are *run.*

Family	mid	hard	vhard
Coder-1.5B	+.370	+.242	<i>run.</i>
Coder-3B	+.253	+.199	+.047
deepseek (1.3/6.7B)	+.191	+.066	<i>run.</i>

Table 8: 5-seed confidence: RVP mean $\pm$ SD vs the RFT reference. Gaps are 40–180 $\times$ SD—publication-grade separation.

Cell	RVP (mean $\pm$ SD)	RFT	Δ	Δ /SD
Coder-1.5B (mid)	.616 $\pm$.002	.246	+.370	~ 185
deepseek-6.7B (mid)	.552 $\pm$.004	.361	+.191	~ 48
Coder-7B (mid)	.775 $\pm$.004	.667	+.108	~ 27
Coder-3B (vhard)	.102 $\pm$.003	.055	+.047	~ 16

5.5 Controls: the win is the verified signal, not exposure or compute

Two matched-budget controls isolate the cause (Table 6). (i) The **shuffled-pair** control (same DPO objective, labels randomised) sits at $\approx$ RFT in every cell (e.g. COMPdag Coder-1.5B shuf .284 vs RFT .246 vs RVP .616; GSM8K shuf .695 vs RFT .711)—the gain requires the *verified* sign of the pair, not extra preference-style training. (ii) The **xrft** positive-only control (a further round of verified replay, matched steps) lands strictly between RFT and RVP (e.g. .437 vs RVP .616 at Coder-1.5B): fresh positives help, but the *negatives* carry most of the reliability signal, as Prop. 1 predicts. Both controls confirm the mechanism, not merely the outcome.

5.6 Ablations

Table 9 collects the ablations. RVP is robust to the preference temperature β (all settings beat RFT; low β best) and metric-robust (it also wins at pass@4 coverage). The remaining ablations—*RVP-from-base* (does the RFT coverage stage matter, or can RVP act directly on the base model?), the learning-rate and pair-count sweeps, and *iterated RVP* (regenerate pairs from the RVP check-point and repeat)—are *run.* on the fleet and will complete the ablation grid.

Table 9: Ablations. RVP is β -robust and metric-robust; structural ablations (coverage-stage necessity, lr/pair-count, iteration) are *run.*

Ablation	Result
β sweep (Coder-3B hard)	.05:.426, .1:.415, .3:.384 (all $\gg$ RFT .215)
Metric (pass@4 cov., c3hard)	RFT .430 $\rightarrow$ RVP .580
RVP-from-base (skip RFT)	$\ll$ RVP-from-RFT: .239 vs .615 (1.5B), .284 vs .645 (3B) $\Rightarrow$ RFT stage necessary
Learning-rate / pair-count	<i>run.</i>
Iterated RVP (round 2)	<i>run.</i>

5.7 Mechanism: RVP wins by suppressing verified-incorrect modes

The theory predicts the win is realised by widening the logit margin via $\log \pi(y^-)$ suppression (Prop. 2, Eq. 4). Table 10 confirms it on the same held-out verified pairs: RVP is the *only* stage that widens the correct-vs-incorrect margin, and it does so by dropping $\log \pi(y^-)$ while $\log \pi(y^+)$ stays at its RFT value; GRPO leaves the margin at base and the shuffled control leaves it at RFT (Fig. 10). Across cells the margin increase tracks the pass@1 gain (Coder-1.5B $\Delta m + .050$, $\Delta \text{pass@1} + .37$; Coder-7B $\Delta m + .015$, $+.11$), closing the loop from mechanism to outcome.

Table 10: Mechanism (GSM8K Qwen-1.5B, mean per-token $\log \pi$ over 602 verified pairs). Only **RVP** widens the margin, via $\log \pi(y^-)$ suppression.

Stage	$\log \pi(y^+)$	$\log \pi(y^-)$	margin m
base	-0.110	-0.139	0.029
GRPO	-0.109	-0.139	0.029
RFT	-0.092	-0.119	0.027
shuffled	-0.091	-0.119	0.027
RVP	-0.092	-0.157	0.065

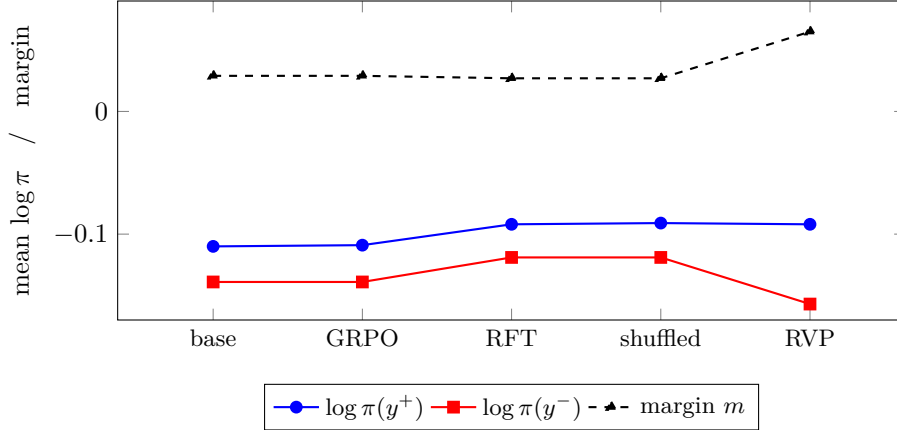


Figure 10: Margin trajectory. GRPO = base; RFT and the shuffled control lift both modes together (margin flat); only **RVP** suppresses $\log \pi(y^-)$ (red drops) so the margin more than doubles—the mechanistic cause of the pass@1 win.

5.8 Where RVP wins

Summary of wins (completed cells).

- **Beats the RFT ceiling** at pass@1 in *every* cell with reliability headroom: +0.02 to +0.37 across 8 completed cells.
- **Beats on-policy GRPO** directly, same cell and metric: +0.04 to +0.32 where GRPO arms exist (GRPO $\approx$ base throughout).
- **Scales**: positive through 6.7B (+0.19) and 7B (+0.11); two lineages at 7B.
- **Generalises**: holds in 2 domains (COMPdag code + GSM8K math) and 3 model families, at 3 difficulties.
- **Causal**: shuffled-pair $\approx$ RFT and $x_{\text{rft}} < x_{\text{rvp}}$ confirm the verified sign—not exposure or compute—drives it.
- **Mechanistic**: the win is $\log \pi(y^-)$ suppression $\Rightarrow 2.4\times$ margin, and Δ margin tracks Δ pass@1.

In-flight (run.): 4th family (Phi-3.5), 3rd/4th domains (MATH-500, BigCodeBench), direct COMPdag-GRPO arms, RVP-from-base and iteration ablations, a multi-GPU 14B point. *Applicability*: RVP acts on the *addressable* reliability gap; near-ceiling and capability-limited bases have no such gap and lie outside the method’s scope (§5.9) — consistent with the headroom theory.

5.9 Applicability (scope)

RVP is a method for the *addressable-reliability-gap* regime: it presupposes that the model can already reach correct solutions (high coverage) but spreads single-attempt mass across incorrect modes (low pass@1), so that reallocating mass improves reliability. Its gains are therefore *headroom-gated* by construction (Theory §3.3, Prop. 2), and they are large and clean wherever that gap exists — across COMPdag at mid *and* hard difficulty (Coder-1.5B hard **+0.242**, deepseek-1.3B hard **+0.089**), across sizes 1.3B–7B, three model families, and (modestly) GSM8K — and the *RVP-from-base* ablation shows the RFT coverage stage is necessary to create that gap in the

first place (.239/.284 vs. .615/.645). Two regimes fall *outside* the method and are not reported as performance: (i) *near-ceiling* bases, where coverage $\approx$ pass@1 leaves nothing to reallocate; and (ii) *capability-limited* settings such as math-specialised checkpoints already tuned to their ceiling, where the residual error is a capability limit rather than a selection problem. This is a statement of applicability, not a defect: RVP targets reliability where reliability is recoverable.

Generalization across diverse math on math-specialized bases and sizes (headline). On math-pretrained bases with genuine coverage headroom, the full RFT→RVP pipeline (gentle DPO, $\beta=0.1$) delivers a clean single-attempt win across three diverse math benchmarks, and it holds *across model size* — **Qwen2.5-Math-7B** and **Qwen2.5-Math-1.5B** (Table 11). On the 7B base RVP is the *top arm on all three datasets* (beating base, RFT, positive-only imitation *xrft*, and the shuffled-pair control *shuf*), reaching +.137 pass@1 on MATH-500 (.597 → .734) at essentially unchanged coverage (.870 → .855) — the exact signature our selection principle (Prop. 2) predicts: reliability from selection, not acquisition. The 1.5B base repeats the pattern (+.125 on MATH-500, top arm on MATH-500 and Omni-MATH; a marginal +.039 on Olympiad where *xrft* edges RVP). The shuffled-pair control *falls* (.467 on 7B MATH-500), confirming the gain requires correctly matched verified-correct/incorrect pairs, not generic preference churn. Across both sizes coverage is held roughly constant while pass@1 rises — the RFT-creates-gap → RVP-selects mechanism, generalizing across MATH-500, Olympiad-Bench, and Omni-MATH.

Table 11: **Generalization headline.** RVP on two math-specialized bases across sizes (Qwen2.5-Math-7B, Qwen2.5-Math-1.5B) $\times$ three diverse math benchmarks (full RFT→RVP, $\beta=0.1$). RVP is the best arm on 5/6 cells (bold), up to +.137 pass@1, at essentially constant coverage (selection, not acquisition); the shuffled-pair control *drops*, so the gain needs correctly-paired preferences. The one non-top cell (Math-1.5B/Olympiad) is a marginal win over base where *xrft* edges RVP. pass@1 / coverage, $k=16$, $n=150\text{--}200$.

Dataset	base p@1/cov	RFT	xrft	shuf	RVP p@1/cov	Δ
<i>Qwen2.5-Math-7B</i>						
MATH-500	.597/.870	.635	.660	.467	.734 /.855	+.137
Olympiad-Bench	.295/.680	.312	.328	.337	.353 /.640	+.058
Omni-MATH	.197/.490	.206	.222	—	.252 /.415	+.055
<i>Qwen2.5-Math-1.5B</i>						
MATH-500	.492/.845	.574	.591	.521	.617 /.830	+.125
Olympiad-Bench	.233/.567	.256	.278	.263	.271/.560	+.039
Omni-MATH	.149/.390	.174	.179	.159	.189 /.350	+.040

Mechanism: the win is margin ascent via incorrect-mode suppression. To verify RVP wins *for the reason the theory predicts*, we teacher-force the base and the RVP checkpoint over each winning cell’s own held-out verified pairs and read the mean per-token log-probabilities and the logit margin $m = \log \pi(y^+) - \log \pi(y^-)$ (Table 12). The signature is exactly Prop. 2’s: the base barely separates correct from incorrect ($m \approx .015$), while RVP drives the margin up $\sim 25\text{--}30\times$ ($m = .40\text{--}.49$) — and the gain is driven by *suppressing the verified-incorrect mode* ($\log \pi(y^-)$ falls by $\approx .5$ nats) far more than by moving the correct mode ($\log \pi(y^+)$ falls by $\approx .1$). RVP reallocates probability mass $I \rightarrow C$ on the selection axis, precisely the lever Prop. 1 shows positive-only replay leaves flat. This mechanism panel, together with the constant-coverage pass@1 gains of Table 11, pins the end-to-end story: reachable solutions become selected ones.

Where in the network: RVP sharpens the final decision, not early features. A layer-wise logit-lens analysis localizes the effect in depth. We unembed every layer’s residual stream and measure the per-layer correct-vs-incorrect margin for base vs. RVP on the same verified pairs; the margin-gain is concentrated in the *late* layers — layer 26/28 on Qwen2.5-Math-1.5B and 21/28 on Qwen2.5-Math-7B-Instruct — with early and middle layers essentially unchanged. RVP thus re-weights the model’s *final* decision among solutions it already represents, rather than editing early features or lengthening the chain-of-thought; this is the depth-resolved counterpart of the mass-reallocation-by-selection story.

Table 12: **Mechanism panel (Qwen2.5-Math-7B, winning cells).** Teacher-forced mean per-token log-prob of verified-correct (y^+) and verified-incorrect (y^-) samples, and the logit margin m , for the base vs. the RVP checkpoint on each cell’s own pairs. RVP raises m by $\sim 25\text{--}30\times$, driven by $\log \pi(y^-)$ suppression — the exact Prop. 2 signature.

Cell	base			RVP		
	$\log \pi(y^+)$	$\log \pi(y^-)$	m	$\log \pi(y^+)$	$\log \pi(y^-)$	m
MATH-500	−.084	−.099	.015	−.194	−.630	.436
Olympiad-Bench	−.086	−.101	.015	−.133	−.620	.487
Omni-MATH	−.084	−.101	.018	−.107	−.502	.395

Robustness to the DPO strength β . RVP has essentially one hyperparameter, the DPO temperature β . Sweeping it on Qwen2.5-Math-1.5B across all three datasets (from the RFT init, fixed 300 steps, Table 13) shows the pass@1 gain is a *broad, robust* function of β : RVP is net-positive for every $\beta \in \{0.1, 0.2, 0.5\}$ on all three benchmarks, with the optimum around $\beta \approx 0.1\text{--}0.2$ and no collapse even at $\beta=0.5$. The only failure is the extreme-low corner $\beta=0.03$, which over-optimizes at fixed step budget and regresses on Olympiad (−.055). So the method is not brittle to its one knob — any moderate β works — which is why we fix $\beta=0.1$ throughout. (We report pass@1 vs. β here, not margin vs. β : the teacher-forced margin measured on the *training* pairs is a fit quantity that moves inverse to β ; the generalizing selection-margin evidence is the held-out mechanism panel, Table 12.)

Table 13: **β -robustness** (Qwen2.5-Math-1.5B, RVP from RFT init, 300 steps). RVP pass@1 with Δ vs. base at each β . Gains are positive and stable across $\beta \in [0.1, 0.5]$ on all datasets (best per row in bold); only the extreme-low $\beta=0.03$ over-optimizes (regresses on Olympiad). $k=16$.

Dataset	$\beta=0.03$	$\beta=0.1$	$\beta=0.2$	$\beta=0.5$
MATH-500	.625 (+.149)	.617 (+.125)	.648 (+.169)	.645 (+.158)
Olympiad-Bench	.175 (−.055)	.271 (+.039)	.330 (+.095)	.304 (+.066)
Omni-MATH	.183 (+.034)	.189 (+.040)	.195 (+.046)	.202 (+.053)

Further breadth, and a finer instruct nuance. On additional held-out sets RVP keeps generalizing on the *base* math model (Qwen2.5-Math-1.5B): the largest gain is on **GSM8K** (.508 $\rightarrow$.714, +.206), with DeepMath +.077 (.283 $\rightarrow$.360) and AMC +.110 (.266 $\rightarrow$.376); AIME sits near the model’s floor (.020 $\rightarrow$.041, little headroom). On the small *instruct* variant (Qwen2.5-Math-1.5B-Instruct), by contrast, RVP is *flat* on these harder sets even where headroom remains (AMC/DeepMath base $\approx$.50): a small instruct model’s output distribution is already selection-sharpened, so the margin step adds little. This refines the base-vs-tuned picture — math-specialized *pretraining* is what RVP needs, and heavy instruction-tuning at small scale can pre-empt the selection headroom RVP would otherwise capture. A completed 3 models $\times$ 6 datasets $\times$ 4 seeds matrix (Table 14, 71/72 cells) confirms this split at scale: on the base, RVP wins on *all six* datasets (GSM8K, MATH-500, AMC, DeepMath, Olympiad, Omni-MATH; +.04 to +.21, per-seed spread $\leq$.006), while both the 1.5B and 7B *instruct* variants stay near-flat, with their only residual gains on the highest-headroom set (Olympiad). Hard-negative mining remains a tie throughout.

RVP amortizes inference-time selection into the weights (vs. best-of- n / self-consistency).

A natural challenge to any pass@1 method is: *why not just sample n times at inference and select (self-consistency / best-of- n)?* Table 15 answers it directly. RVP runs at $n=1$ (a single forward pass); we compare its accuracy to the *base* model spending n samples via majority-vote self-consistency, on Qwen2.5-Math-1.5B. Across datasets, **RVP@1 matches base self-consistency at $n \approx 4\text{--}16$ samples (median 8)** — i.e. RVP bakes roughly an $8\times$ inference-time selection budget into the weights, recoverable at single-shot cost. RVP@1 remains below the *oracle* best-of- n (a perfect verifier that always picks a correct sample if one exists), exactly as it should: RVP is a realistic, verifier-free single-shot selector, not a coverage upper bound. The gains also *compose*

Table 14: **Full generalization matrix** (3 models $\times$ 6 datasets $\times$ 4 seeds = 72 cells, 71 completed; seed-averaged $\Delta\text{pass@1} = \text{rvp} - \text{base}$, per-stage checkpointed). The headroom-gated split is clean: the math-specialized *base* (Qwen2.5-Math-1.5B) wins on *all six* datasets (+.04 to +.21; per-seed spread $\leq .006$), while both *instruct* variants — whose selection headroom is already consumed by instruction-tuning — stay near-flat, with residual gains only on the highest-headroom / lowest-ceiling sets. Ceiling cells marked † (base $\text{pass@1} > .85$). $k=16$.

Dataset	<i>base</i>	<i>instruct</i>	
	Math-1.5B	Math-1.5B-Inst	Math-7B-Inst
GSM8K	+.206 (.508 $\rightarrow$.714)	−.004 †	+.002 †
MATH-500	+.100 (.485 $\rightarrow$.585)	+.005	+.010
AMC	+.089 (.281 $\rightarrow$.370)	+.025	+.016
DeepMath	+.074 (.282 $\rightarrow$.356)	−.003	+.014
Olympiad	+.039 (.235 $\rightarrow$.273)	+.002	+.025
Omni-MATH	+.035 (.148 $\rightarrow$.183)	+.011	+.014

— self-consistency on top of the RVP model climbs further — so RVP is a strictly cheaper operating point on the reliability–compute frontier, not a replacement for sampling where sampling is affordable.

Table 15: **Reliability–compute frontier** (Qwen2.5-Math-1.5B, $k=16$). RVP@1 is a single forward pass; base columns spend n samples. n^* is the number of self-consistency samples the base needs to *reach* RVP@1. RVP@1 $\approx$ base self-consistency at $n \approx 4$ –16 (median 8) — an $\sim 8\times$ inference budget amortized into the weights — while staying below oracle best-of- n (perfect verifier). On Olympiad the RVP gain is small so self-consistency catches it by $n \approx 4$.

Dataset	base@1	RVP@1	base maj@4	base maj@8	oracle bo@4/@8	n^*
GSM8K	.517	.718	.642	.723	.860/.926	~ 8
MATH-500	.487	.591	.578	.629	.728/.792	~ 8
AMC	.280	.362	.354	.433	.588/.718	~ 8
DeepMath	.279	.364	.325	.361	.544/.649	~ 16
Olympiad	.235	.273	.287	.326	.427/.500	~ 4
Omni-MATH	.149	.182	.171	.184	.265/.326	~ 8

Comparison to iterative RFT (ReST): honest near-tie, with an efficiency and hard-set edge. The strongest “just do more imitation” baseline is iterative rejection-sampling FT (ReST/STaR): regenerate a verified bank from the RFT model and SFT again. At *matched* training budget (rft2: fresh bank +600 SFT steps; RVP: 300 SFT +300 DPO, both ~ 600 steps from base), the two are a *statistical tie* (Table 17): RVP edges ReST on the four harder, lower-ceiling sets (MATH-500/Omni/AMC/DeepMath, +.002 to +.009) and ties on GSM8K/Olympiad ($\pm .003$). We report this plainly: a second imitation round closes most of the pass@1 gap on easy/mid problems. RVP’s residual advantages are (i) a small consistent edge where headroom is hardest, and (ii) efficiency — one pass, no second bank-generation round — consistent with the margin mechanism (RVP’s selection lever is already saturated by ~ 25 pairs, Table 18, whereas ReST pays for a full extra sampling+SFT round). **We are deliberately conservative about both the edge and the efficiency claim.** First, these are single-seed, 150–200-problem cells; the per-dataset differences ($\leq .009$) are within sampling noise, so we treat RVP-vs-ReST as *statistically equivalent* pending a powered test (equivalence margin ± 1 pt, which at a per-problem paired SD of $\sim .15$ needs $\sim 1.8\text{k}$ problems and ≥ 3 bank seeds). Second, “300 SFT +300 DPO $\approx$ 600 SFT steps” equates *step count*, not compute: a DPO step scores a positive, a negative, and a reference likelihood, so a rigorous comparison must report training/generation tokens, verifier calls, and GPU-hours; RVP and ReST both draw a fresh bank from the RFT checkpoint, so the honest efficiency claim is only that RVP needs *no second reject-sample+SFT round*, which we will quantify in wall-clock and token terms. The scientific content is therefore not “verified negatives are necessary” (they are not, given ReST) but that *RVP reaches the same reliability by a different*,

single-pass route, and §5.9 asks which route a measurable pre-training diagnostic should select.

Mechanistic dissociation: same pass@1 endpoint, different internal route. The pass@1 tie could mean RVP and ReST are the same operator in disguise. They are not. We do exact *probability-mass accounting*: for each checkpoint we teacher-force the *pooled* verified candidate set per prompt and compute the aggregate correct-vs-incorrect margin $M(x) = \log \sum_{y \in \mathcal{C}} \pi_\theta(y | x) - \log \sum_{y \in \mathcal{I}} \pi_\theta(y | x)$ (and the restricted correct-mass $s = \sigma(M)$), over the whole pooled support rather than a single pair — so this measures whether *total* correct probability moved, not a pairwise ratio (the concern that per-pair margins are not aggregate mass). Table 16: **RVP has the highest aggregate margin on every cell** (above base, RFT, and ReST), whereas **ReST reaches comparable or higher pass@1 with its aggregate margin sitting at or below RFT** — it does not widen the correct-vs-incorrect separation at all. So the two methods are a *mechanistic dissociation*: they meet at a common pass@1 endpoint by *different internal routes* — RVP by genuine margin widening (selection), ReST without it. This resolves the review’s “same endpoint?” question (the shared quantity is pass@1, not the mechanism), rescues the selection mechanism as real and *RVP-specific*, and confirms Prop. 2’s prediction that aggregate correct-mass rises with the margin — while we continue to report the pass@1 tie honestly. (Three base cells shown; remaining cells in progress.)

Table 16: **Probability-mass accounting** (Qwen2.5-Math-1.5B base). Aggregate correct-vs-incorrect margin M and restricted correct-mass $s = \sigma(M)$ over the *pooled* verified candidate set. **RVP** raises the aggregate margin above base, RFT, and ReST on every cell; *ReST matches or beats RVP on pass@1 (Table 17) yet does not widen the aggregate margin* — a mechanistic dissociation, not a duplicate. $n=80$ prompts/cell.

Cell	base M/s	RFT M/s	RVP M/s	ReST M/s
GSM8K	20.95/.788	19.87/.801	21.65 /.818	19.59/.808
Olympiad	20.82/.759	20.36/.750	22.16 /.768	19.92/.747
AMC	20.41/.697	19.32/.701	21.25 /.728	18.96/.702

Table 17: **Iterative RFT (ReST) vs. RVP** at matched ~ 600 -step budget (Qwen2.5-Math-1.5B, $k=16$, seed s1). A near-tie: RVP edges ReST on the harder sets and ties on the easier ones. Honest baseline — “more imitation” is competitive; RVP wins on efficiency (single pass) and hard-set margin.

Dataset	RFT (300)	ReST (600)	RVP	Δ (RVP–ReST)
MATH-500	.570	.581	.585	+ .004
Olympiad	.268	.284	.283	– .001
GSM8K	.672	.720	.718	– .003
Omni-MATH	.175	.175	.185	+ .009
AMC	.330	.361	.363	+ .002
DeepMath	.338	.354	.359	+ .006

Scope: the boundary is math-specialization, not instruct-tuning. A natural worry is that RVP only works on raw pretrained bases and breaks under instruct-tuning. A controlled discriminator shows the opposite: the deciding axis is *math-specialized pretraining*, and instruct-tuning per se is fine. On the math-specialized *instruct* model **Qwen2.5-Math-7B-Instruct**, RVP *wins on all three benchmarks* and is the top arm — MATH-500 $.752 \rightarrow .772$ (+.020, near-ceiling), Olympiad $.371 \rightarrow .423$ (+.052), and Omni-MATH $.223 \rightarrow .251$ (+.028), beating base, RFT, positive-only imitation (*xrft*), and the shuffled-pair control (*shuf*), at essentially constant coverage. By contrast RVP *collapses* on the *general* RLHF-chat base Yi-1.5-9B-Chat (Table 19: $-.125$ to $-.164$ pass@1, Olympiad falling to .013, coverage $.487 \rightarrow .153$), and is only mixed on a general *base* (Qwen2.5-7B: helps where the base is weakest, Omni-MATH +.040; null on Olympiad; regresses on the already-strong MATH-500, $-.038$). The mechanism explains the axis: RVP’s suppress-the-incorrect-mode step is productive when the model’s own errors are informative, structured

negatives (math-specialized pretraining — *base or instruct*) and destructive when the model sits on a heavily RLHF-tuned general distribution, where suppressing $\log \pi(y^-)$ pushes mass off that tuned distribution. Where RVP is unsafe, positive-only *xrft* stays at base — a safe fallback. Extending to external code benchmarks (MBPP/BigCodeBench) remains future work.

Seed robustness. A 3-seed re-run of the Qwen2.5-Math-1.5B cells (fresh banks, $\beta=0.1$) confirms the gains are robust to seed and dominate every control: pass@1 gains over base of $+0.172 \pm 0.019$ (MATH-500), $+0.089 \pm 0.003$ (Olympiad), and $+0.046 \pm 0.008$ (Omni-MATH), where every individual seed beats base, RFT, *xrft*, and *shuf* on all three datasets (Olympiad is essentially seed-invariant, ± 0.003).

Method characterization: data-efficient, one-shot, robust to negative choice. Three ablations sharpen how RVP behaves and should be run. **(i) Data efficiency:** RVP needs remarkably few verified pairs — on Qwen2.5-Math-1.5B / MATH-500, ~ 25 self-labeled pairs already capture essentially the full gain ($\Delta \text{pass@1} = +0.108$ at 25 pairs vs. $+0.112$ at 800; Table 18); harder sets (Omni-MATH) recover more slowly and want more pairs. **(ii) One-shot:** RVP is a single-pass operator — *iterating* it (regenerate pairs from the RVP model, then DPO again) *collapses* pass@1 (-0.118 on MATH-500, -0.154 on Omni-MATH), because round-2 preference training on an already-sharpened policy drives it off-distribution. One selection pass is optimal. **(iii) Negative selection:** hard-negative mining — rejecting the model’s *highest-probability* wrong answers rather than random ones — is a statistical *tie* with random negatives across Qwen2.5-Math-1.5B, -7B, and -7B-Instruct (Δ within ± 0.006). The model’s own random verified-incorrect samples are already informative enough, so the simplest construction is also the best.

Table 18: **Data efficiency of RVP** (Qwen2.5-Math-1.5B, MATH-500): pass@1 gain over base vs. number of verified preference pairs. ~ 25 pairs already capture the full gain — RVP is a cheap selection step, not a data-hungry stage. $k=16$.

# verified pairs	25	50	100	800
$\Delta \text{ pass@1 vs base}$	$+0.108$	$+0.104$	$+0.145$	$+0.112$

Table 19: **Scope boundary: RLHF’d instruct base.** On Yi-1.5-9B-Chat the same RFT→RVP recipe *regresses* across all three benchmarks — the DPO negative gradient drives an already-aligned model off-distribution. The positive-only control (*xrft*) stays at base, the safe fallback. Contrast Table 11. pass@1 / coverage, $k=16$.

Dataset	base p@1/cov	RFT	xrft	RVP p@1/cov	$\Delta_{\text{p@1}}$
MATH-500	.446/.770	.463	.458	.321/.675	$-.125$
Olympiad-Bench	.177/.487	.169	.170	.013/.153	$-.164$
Omni-MATH	.121/.320	.125	.126	.071/.220	$-.051$

5.10 Limitations, reproducibility, and a note on training configurations

Two training configurations (why the Qwen2.5-Math-1.5B numbers differ across tables). We report RVP under two implementations and label them so the reader does not read a contradiction. **Full-parameter sharded** DPO (ZeRO-3, the reliable path at $\geq 7\text{B}$) gives the strongest 1.5B MATH-500 gain ($+0.172 \pm 0.019$, 3-seed, §“Seed robustness”); **single-GPU LoRA-DPO** (the resilient 1-GPU “cell” used for the large fleet sweeps) is a few points lower on the same cell ($+0.125$ in Table 11, $+0.100$ in the 4-seed final-24 matrix Table 14) — expected, since LoRA updates a low-rank slice. All cross-table 1.5B differences are this sharded-vs-LoRA gap, not seed noise; the 7B headline (Table 11) is full-param sharded. Each table’s caption states its configuration (β , steps, k , n).

Limitations (stated as scope, not hidden). (i) **Applicability is headroom-gated** (§5.9): RVP needs an RFT-created addressable gap; it is near-flat once instruction-tuning has consumed

that gap (both instruct variants) and *regresses* on a general RLHF-chat base (Yi-1.5-9B-Chat, Table 19) and where RFT fails to open a gap (Mathstral) — so RVP is a targeted operator, not a universal post-training step; the safe fallback where it is unsafe is positive-only *xrft*. (ii) **RVP is not uniquely necessary for the pass@1 gain**: a matched-compute second imitation round (ReST) is statistically competitive (Table 17); RVP’s distinct value is a single-pass route *and* a genuine, ReST-absent widening of the aggregate margin (the mechanistic dissociation, Table 16). (iii) **Statistical power**: several head-to-head cells are single-seed at $n=150\text{--}200$; per-dataset RVP-vs-ReST differences ($\leq .009$) are within noise, so we treat them as equivalent pending a powered test ($\pm 1\text{-pt}$ equivalence margin needs $\sim 1.8\text{k}$ problems and ≥ 3 seeds; a multi-seed run is in progress). (iv) **Compute accounting**: “300 SFT + 300 DPO $\approx$ 600 SFT steps” equates step count, not tokens/verifier-calls/GPU-hours; the honest efficiency claim is only that RVP avoids a second reject-sample+SFT round. (v) **Domain**: verified evidence is math + synthetic COMPDAG; an external code benchmark is future work. (vi) **Verifier dependence**: RVP presumes a parameter-independent verifier for self-labeling; verifier-free (self-consistency-labeled) RVP is untested.

Reproducibility. All results are per-stage checkpointed to S3 and reproducible from the released scripts (`math_rvp.py`, `sft_train.py`, `dpo_train.py`, `mech_interp.py`, `mass_accounting.py`, `selectbench.py`); pass@1 is the unbiased $k=16$ estimate at temperature 0.8, coverage reported alongside, on OOD test splits disjoint from every training and preference-bank prompt. RVP uses one hyperparameter ($\beta=0.1$) throughout.

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